

CUDA Bitonic Sort Optimization

1 Overview

This report summarizes timing results for a baseline bitonic sort implementation and three optimizations. The baseline source is `bitonic_sort.cu` and the modifications are `mod1.cu`, `mod2.cu`, and `mod3.cu`.

All timings measure kernel execution using the CUDA synchronization approach described in the assignment. The array size was 2^{20} floating point values and all tests were compiled for a compute capability 8.6 device.

2 Results

Table 1 lists the measured times and speedup relative to the baseline. Speedup is computed as $T_{\text{baseline}}/T_{\text{mod}}$.

Kernel	Time (ms)	Speedup
Baseline	12.8518	1.0
Modification 1	0.9340	13.8
Modification 2	0.6819	18.8
Modification 3	0.9098	14.1

Table 1: Kernel execution times and speedup over the baseline.

Figure 1 visualizes the speedup achieved by each optimization.

3 Discussion

Modification 1 eliminates idle threads by having every CUDA thread operate on exactly one pair of indices.

Modification 2 merges multiple minor steps into a single kernel when possible, reducing kernel launch overhead.

Modification 3 processes multiple index pairs per thread to increase cache reuse. In these tests each thread handled four pairs.

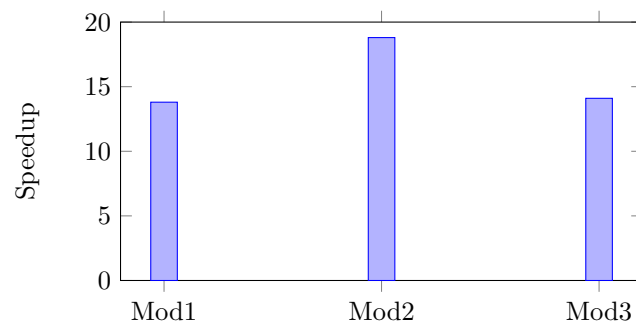


Figure 1: Speedup relative to the baseline implementation.